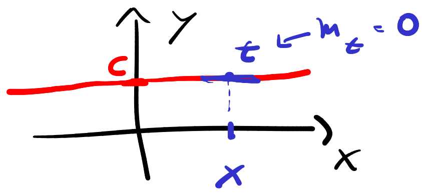
$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$\updownarrow$

derivada de f no ponto x $\}$ $\longrightarrow$ Derivada no ponto x é um limite (pode existir ou não)

//

Exemplos:



① $f(x) = c$; $c \in \mathbb{R}$

$$f'(x) = \lim_{h \to 0} \frac{\overbrace{f(x+h)}^{c} - \overbrace{f(x)}^{c}}{h}$$

$$= \lim_{h \to 0} \frac{0 - 0}{h} = \lim_{h \to 0} 0$$

$$= 0$$

DERIVADA DE FUNÇÃO CONSTANTE NO PONTO $x \in D_f$ É 0.

② $f(x) = ax$; $a \in \mathbb{R}$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{a.(x+h) - ax}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{ax} + a\cancel{h} - \cancel{ax}}{\cancel{h}}$$

$$= \lim_{h \to 0} a = \underline{\underline{a}}$$

$$f(x) = ax \implies f'(x) = a$$

3) $f(x) = x^2 \implies \underline{\underline{f'(x)}} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \ldots = \underline{\underline{2x}}$

4) $f(x) = \sqrt{x} \implies f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \ldots$

TÉCNICA DO CONJUGADO

$$= \frac{1}{2\sqrt{x}} = \frac{1}{2} x^{-1/2}$$

$$\begin{cases} \cos(a \pm b) = \cos a . \cos b \mp \operatorname{sen} a . \operatorname{sen} b \\ \operatorname{sen}(a \pm b) = \operatorname{sen} a \cos b \pm \operatorname{sen} b \cos a \end{cases}$$

5) $f(x) = \operatorname{sen} x$

$$\operatorname{sen}^2 a + \cos^2 a = 1$$

(RFT)

LIMITE TRIGONOMÉTRICO FUNDAMENTAL $\left\{ \lim_{h \to 0} \frac{\operatorname{sen} h}{h} = 1 \right.$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\operatorname{sen}(x+h) - \operatorname{sen} x}{h}$$

$$= \lim_{h \to 0} \frac{\operatorname{sen} x \cos h + \operatorname{sen} h \cos x - \operatorname{sen} x}{h}$$

$$= \lim_{h\to 0}\left(\frac{\cos x \cdot \operatorname{sen} h}{h} + \underbrace{\frac{\operatorname{sen} x \cdot \cos h}{h} - \frac{\operatorname{sen} x}{h}}_{\frac{-\operatorname{sen} x\,(-\cos h + 1)}{h}}\right)$$

$$= \lim_{h\to 0}\left(\frac{\cos x \operatorname{sen} h}{h} - \operatorname{sen} x\left(\frac{1-\cos h}{h}\right)\right)$$

constante

constante

$$= \lim_{h\to 0}\left(\frac{\cos x \cdot \operatorname{sen} h}{h}\right) - \lim_{h\to 0} \operatorname{sen} x \left(\frac{1-\cos h}{h}\right)$$

$$= \cos x \cdot \underbrace{\lim_{h\to 0}\left(\frac{\operatorname{sen} h}{h}\right)}_{1} - \operatorname{sen} x \cdot \underbrace{\lim_{h\to 0}\left(\frac{1-\cos h}{h}\right)}_{\text{Ex. 3.2 Problema 3.16} \;=\; 0}$$

$$= \cos x \cdot 1 - \operatorname{sen} x \cdot 0$$

$$= \cos x$$

$$f(x) = \operatorname{sen} x \Rightarrow f'(x) = \cos x$$

(6) $f(x) = \cos x \Rightarrow f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$

$$= \ldots = -\operatorname{sen} x$$

⑦ $f(x) = x^n \; ; \; n \in \mathbb{N}$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$(x+h)^n = \cancel{x^n} + \underbrace{\binom{n}{1} x^{n-1} \cdot \cancel{h}}_{n \cdot x^{n-1}} + \binom{n}{2} x^{n-2} \cdot h^2 + \dots + h^n$$

$$= \lim_{h \to 0} \frac{(x+h)^n - \cancel{x^n}}{\cancel{h}}$$

$$= n \cdot x^{n-1}$$

$$f(x) = x^n, \; n \in \mathbb{N} \Rightarrow f'(x) = n \cdot x^{n-1}$$

⑧ $f(x) = x^n \; ; \; \underline{\underline{n \in \mathbb{R}}} \Rightarrow f'(x) = n \cdot x^{n-1}$

$$n = \sqrt{2} \Rightarrow f'(x) = \sqrt{2} \cdot x^{\sqrt{2}-1}$$

$$f(x) = x^{\sqrt{2}}$$

⑨ $f(x) = a^x \; ; \; a \in \underbrace{\mathbb{R}^*_+}_{a > 0}$

$$a^0 = 1$$

$$a^1 = a$$

$$a^{-m} = \frac{1}{a^m}$$

$$a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

$$a^{m+n} = a^m \cdot a^n$$

$$a^{m-n} = a^{m+(-n)} = a^m \cdot a^{-n} = \frac{a^m}{a^n}$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{a^{x+h} - a^x}{h}$$

$$= \lim_{h\to 0} \frac{a^x \cdot a^h - a^x}{h}$$

$$= \lim_{h\to 0}\left(\boxed{a^x}\,\frac{(a^h - 1)}{h}\right)$$

CONSTANTE

$$= a^x \cdot \lim_{h\to 0}\left(\frac{a^h - 1}{h}\right) = a^x \cdot f'(0)$$

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$$

Note que $\ln(a) = f'(0)$

$$= \lim_{h\to 0}\left(\frac{a^h - 1}{h}\right) \overset{\text{DEF}}{\underset{\text{Def. 4.3}}{=}} \ln(a)$$

Se $f(x) = a^x$, então $\boxed{f'(x) = a^x \cdot f'(0)}$

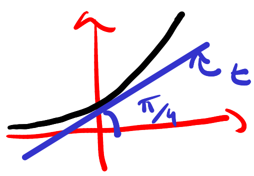
Logo, $\boxed{f'(x) = a^x \cdot \ln a}$

CASO PARTICULAR:

$$a = e \Rightarrow \ln e = 1 \Rightarrow f'(x) = e^x \cdot \overset{1}{\ln e} = e^x$$

$f(x) = e^x \Rightarrow f'(x) = e^x$

$f(x) = a^x \to f'(0) = \lim_{h\to 0} \frac{a^h - 1}{h}$



$t \to m_t = f'(0)$

Def. 4.2: e é o nº que satisfaz o limite: $\lim_{h\to 0} \frac{e^h - 1}{h} = 1$

(10) $f(x) = \ln x \implies f'(x) = 1/x$

$f(x) = \log_a x \implies f'(x) = \dfrac{1}{x \cdot \ln(a)}$

(11)

$f(x) = \dfrac{1}{x^c} = x^{-c}$

$a^{m.n} = (a^m)^n = (a^n)^m$

$$f'(x) = \frac{d}{dx}\left(x^{-c}\right) = -c \cdot x^{-c-1}$$

$$= -c \cdot x^{-1.(c+1)}$$

$$= -c \cdot x^{(c+1).-1}$$

$$= -c \cdot \left(\underbrace{\boxed{x^{c+1}}}_{z}\right)^{-1}$$

$z^{-1} = \dfrac{1}{z}$

$$= -c \cdot \frac{1}{x^{c+1}}$$

$$= \frac{-c}{x^{1+c}}$$

(12) $Tg\,x = \dfrac{\operatorname{Sen} x}{\cos x} \implies (Tg\,x)' = \left(\dfrac{\overbrace{\operatorname{Sen} x}^{f(x)}}{\underbrace{\cos x}_{g(x)}}\right)'$

- Derivada do Quociente

$f(x) = \operatorname{sen} x$

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

$g(x) = \cos x$

$$= \frac{(\operatorname{sen} x)' . \cos x - \operatorname{sen} x . (\cos x)'}{(\cos x)^2}$$

$$= \frac{(\cos x). \cos x - \operatorname{sen} x (-\operatorname{sen} x)}{\cos^2(x)}$$

$$= \frac{\cos^2 x + \operatorname{sen}^2 x}{\cos^2 x} \quad \text{R.F.T}$$

$$= \frac{1}{\cos^2 x} \Bigg\} \sec^2 x \qquad \sec x \overset{DEF}{=} \frac{1}{\cos x}$$

//

EXEMPLOS DE USO DAS REGRAS DE DERIVAÇÃO:

Regras de Derivação

- $(cf(x))' = cf'(x)$
- Derivada da Soma
  $$(f(x) + g(x))' = f'(x) + g'(x)$$
- Derivada do Produto
  $$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$
- Derivada do Quociente
  $$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$
- Regra da Cadeia
  $$(f(g(x))' = (f'(g(x))g'(x)$$

$$(cf(x))' = cf'(x)$$

Ex. ① $f(x) = x^5 - 4x^3 + 6$ $\Big\}$ $\Big(\underset{\uparrow c}{3}\underbrace{(x^5 - 4x^3 + 6)}_{f}\Big)' =$

$c = 3$

$$3.\,(x^5 - 4x^3 + 6)'$$

② $f(x) = \overset{c}{5}\,\text{sen}\,x \Rightarrow f'(x) = (5.\text{sen}\,x)'$

$$= 5\,(\text{sen}\,x)'$$

$$= 5.\,\cos x$$

③ $f(x) = \overset{c}{-10}\cos x$

$$f'(x) = (-10.\cos x)' = -10.\,(\cos x)'$$

$$= -10.\,(-\text{sen}\,x)$$

$$= 10\,\text{sen}\,x$$

- Derivada da Soma

$$(f(x) \pm g(x))' = f'(x) \pm g'(x)$$

$\Big\}$ FUNDAMENTAL P/ DERIVADA DE FUNÇÃO POLINOMIAL

$f(x) = x^5 - 4x^3 \Rightarrow f'(x) = (\underbrace{\boxed{x^5}}_{1^a\ \text{função}} - \underbrace{\boxed{4x^3}}_{2^a\ \text{função}})' = (x^5)' - (4x^3)'$

$$= 5x^{5-1} - 4(x^3)' = 5x^4 - 4.3.x^{3-1}$$

$$= 5x^4 - 12.x^2$$

$$\boxed{f'(x) = 5x^4 - 12x^2}$$

Obs: Note que se tivermos mais de funções, a regra continua válida.

$$(\underbrace{f(x) + g(x)}_{z(x)} + h(x))' = (z(x) + h(x))'$$

$$\overset{\text{REGRA}}{=} \underbrace{z'(x)} + h'(x)$$

$$= f'(x) + g'(x) + h'(x)$$

$$(\underbrace{f_1(x) + f_2(x) + f_3(x)}_{z(x)} + f_4(x))' = (z(x) + f_4(x))'$$

$$\overset{\text{REGRA}}{=} \underbrace{z'(x)} + f_4'(x)$$

$$= f_1'(x) + f_2'(x) + f_3'(x) + f_4'(x)$$

⋮

Por recursividade, a regra vale p/ qualquer quantidade de funções (p/ subtração tbm!)

Ex: DETERMINAR $f'(x)$.

$$f(x) = 5x^3 - 2x^2 + 10x - 15$$

$$f'(x) = \left(5x^3 + (-2x^2) + (10x) + (-15)\right)'$$

$$= (5x^3)' + (-2x^2)' + (10x)' + (-15)'$$

$$= 5.\underbrace{(x^3)'}_{3x^2} - 2.\underbrace{(x^2)'}_{2x} + 10\underbrace{(x)'}_{1} + \underbrace{(-15)'}_{0}$$

$$= 15x^2 - 4x + 10 + 0$$

$$f'(x) = 15x^2 - 4x + 10$$

SENDO $f(x) = 5x^3 - 2x^2 + 10x - 15$, determine a equação da reta tangente ao graf.$_f$ no ponto $(2, f(2))$.

INCLINAÇÃO DA TANGENTE EM $x = 2$:

$$f'(2) = 15.(2)^2 - 4.2 + 10$$
$$= 15.4 - 8 + 10$$
$$= 62$$

$(2, f(2)) \in$ Reta.
$x_0 \quad y_0$

eq. reta: $\frac{\Delta y}{\Delta x} = m \iff \frac{y - y_0}{x - x_0} = m$

$$\frac{y - f(2)}{x - 2} = f'(2) = 62$$

$$\frac{y - f(2)}{x - 2} = 62$$

$f(2) = 5.(2)^3 - 2(2)^2 + 10(2) - 15$
$= 40 - 8 + 20 - 15$
$= 37$

$$y - f(2) = 62.(x - 2)$$

$$y = 62(x - 2) + f(2)$$

$$y = 62x - 124 + f(2)$$

$$y = 62x - 124 + 37$$

Eq. da reta tangente a f em $(2, 37)$ $\leftarrow$ $\boxed{y = 62x - 87}$

Se $f(x) = \operatorname{sen} x$, determine a equação da reta tangente ao graf. f no ponto $(1, \underbrace{\operatorname{sen}(1)}_{0{,}84})$

$\in$ à reta Tangente

INCLINAÇÃO DE RETA TANGENTE NO PONTO X $= f'(x) \xRightarrow{x=1} m = f'(1)$

$$f'(x) = (\text{sen}(x))' = \cos(x)$$

$$m = \cos(1)$$

$$\boxed{m = 0{,}54}$$

$$\text{sen}^2 x + \cos^2 x = 1$$

$$\text{sen}^2(1) + \cos^2(1) = 1$$

$$(0{,}84)^2 + \cos^2(1) = 1$$

LOGO, $\dfrac{y - y_0}{x - x_0} = m$

$$\frac{y - 0{,}84}{x - 1} = \frac{0{,}54}{1}$$

$$y - 0{,}84 = 0{,}54(x - 1)$$

$$y - 0{,}84 = 0{,}54x - 0{,}54$$

$$\boxed{y = 0{,}54x + 0{,}3}$$